

HYDROGENATED NBR (HNBR) SYNTHESIS AND PROCESS CONTROL: PATENT LANDSCAPE REVIEW

1. ABSTRACT

This report synthesizes a set of patents relevant to hydrogenated nitrile butadiene rubber (HNBR), with emphasis on how NBR precursors are transformed into hydrogenated products while preserving polymer integrity and controlling residual unsaturation, molecular weight, and catalyst residues. Across the landscape, the core target is HNBR derived from acrylonitrile-butadiene rubber or related nitrile rubbers, typically defined by partial to near-complete hydrogenation of butadiene-derived double bonds, with resulting improvements in heat, oil, ozone, and chemical resistance. The patents fall into two broad technical groups: (1) catalytic hydrogenation of NBR to HNBR, often using ruthenium, rhodium, palladium, or Wilkinson-type systems under elevated hydrogen pressure and temperature; and (2) metathesis-assisted molecular-weight reduction of NBR, sometimes combined with hydrogenation in a one-pot or simultaneous process to produce lower-Mooney, lower-Mw HNBR with narrower molecular-weight distributions. A recurring process theme is the need to reduce residual double bonds while avoiding gel formation, excessive nitrile reduction, or catalyst contamination that degrades aging performance and color. Several patents focus on catalyst selection and loading, including Grubbs-type, Hoveyda-type, Wilkinson's catalyst, and other Ru complexes, while others emphasize post-reaction purification or catalyst recovery to improve product stability. Formulation-oriented patents also show how HNBR is used in conductive pastes and vulcanizable mixtures, where alkali-metal content, filler type, and cure-package selection influence processing and end-use properties. Overall, the evidence indicates that the most effective HNBR synthesis routes combine controlled hydrogenation conditions, careful catalyst choice, and, where needed, pre-hydrogenation metathesis to lower molecular weight without sacrificing polymer microstructure.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis catalyst systems for reacting nitrile rubber in the presence of a metathesis catalyst and one or more salts, with the stated goal of improving catalyst activity in metathesis of nitrile rubber. The process is positioned as a precursor-degradation route relevant to HNBR manufacture because it reduces NBR molecular weight before subsequent hydrogenation. - Salt(s): 0.01:1 to 10000:1 salt:metathesis catalyst weight ratio — Claim 1 - Example 7: 10 mg Grubbs II catalyst — Example 7 - Example 7: 1 mg LiBr — Example 7 - Example 7: 0.3 ml chlorobenzene — Example 7 - Example 7: 0.2 ml CDCl₃ — Example 7 - Example 7: 0.15 ml diethyl diallylmalonate — Example 7 - Example 7: room temperature — Example 7 - Example 7: 2 minutes before monomer addition — Example 7 - Example 8: 10 mg Grubbs II catalyst — Example 8 - Example 8: 1 mg CsBr — Example 8 - Example 8: 0.3 ml chlorobenzene — Example 8 - Example 8: 0.2 ml CDCl₃ — Example 8 - Example 8: 0.15 ml diethyl diallylmalonate — Example 8 - Example 8: room temperature — Example 8

Relevant Experimental Evidence - Example 7: LiBr accelerated ring-closing metathesis of diethyl diallylmalonate relative to the salt-free control. - Example 8: CsBr also accelerated ring-closing metathesis, though less strongly than LiBr in the reported conversion profile.

Technical Relevance Although not a direct hydrogenation patent, it is highly relevant because it addresses metathesis of nitrile rubber, a known upstream step for lowering NBR molecular weight before HNBR formation. The salt-activated catalyst concept is directly applicable to controlling NBR degradation prior to hydrogenation.

Patent 2

- Patent Number: US7737233B2
- Patent Title: Catalyst systems and their use for metathesis reactions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses catalyst systems for metathesis reactions, including nitrile rubber degradation, using ruthenium- or osmium-based metathesis catalysts combined with alkaline earth metal chlorides such as calcium chloride or

magnesium chloride. The process is relevant to HNBR because it supports metathesis-based molecular-weight reduction of NBR before hydrogenation. - Alkaline earth metal chloride: calcium chloride or magnesium chloride — Claim 2 - Catalyst metal: molybdenum, tungsten, osmium or ruthenium — Claim 3 - HNBR precursor metathesis feed Mooney viscosity: 24 to 70 — Description - Preferred NBR feed Mooney viscosity: 28 to 40 — Description - NBR feed Mooney viscosity workable in industrial plant: about 30 Mooney units — Description - Hydrogenated nitrile rubber Mooney viscosity from such feedstock: about 55 Mooney units — Description - Degradation target before hydrogenation: less than 30 Mooney units or $M_n < 70000$ g/mol — Description - Hydrogenated nitrile rubbers from prior art: M_w 30000 to 250000 — Description - Hydrogenated nitrile rubbers from prior art: Mooney viscosity 3 to 50 — Description - Hydrogenated nitrile rubbers from prior art: PDI less than 2.5 — Description - Example catalyst loading in cited prior art: 307 ppm Ru — Description - Example catalyst loading in cited prior art: 61 ppm Ru — Description - Example catalyst loading in cited prior art: 0.5 phr Grubbs I catalyst — Description - Example catalyst loading in cited prior art: 614 ppm Ru — Description

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is relevant because it targets the metathesis step used to precondition NBR for HNBR synthesis, especially by lowering molecular weight and Mooney viscosity before hydrogenation. The salt-additive strategy is important for improving metathesis efficiency and reducing catalyst burden in HNBR precursor processing.

Patent 3

- Patent Number: US9079979B2
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-step process in which nitrile rubber is simultaneously hydrogenated and metathesized in the presence of hydrogen, optionally with a co-olefin, using catalysts capable of both functions. The process is explicitly aimed at producing HNBR with lower molecular weight and narrower molecular-weight distribution than conventional two-step routes. - Hydrogenation catalyst amount: 0.057 phr — HNBR 1, Example 1 - Hydrogenation catalyst amount: 0.080 phr — HNBR 2, Example 1 - Hydrogenation catalyst amount: 0.06 phr — HNBR 3, Example 1 - Hydrogenation catalyst amount: 0.055 phr — HNBR 4, Example 1 - Metal content: 65.6 ppm — HNBR 1, HNBR 3, HNBR 4, Example 1 - Metal content: 91.3 ppm — HNBR 2, Example 1 - Reaction time: 0 min — Example 1 - Reaction time: 60 min — Example 1 - Reaction time: 120 min — Example 1 - Reaction time: 180 min — Example 1 - Reaction time: 240 min — Example 1 - Residual double bond content: 2.0% —

HNBR 1 at 60 min, Example 1 - Residual double bond content: 0.7% — HNBR 2 at 60 min, Example 1 - Residual double bond content: 15.4% — HNBR 3 at 60 min, Example 1 - Residual double bond content: 2.9% — HNBR 4 at 60 min, Example 1 - Residual double bond content: 0.4% — HNBR 1 at 120 min, Example 1 - Residual double bond content: 0.5% — HNBR 2 at 120 min, Example 1 - Residual double bond content: 3.1% — HNBR 3 at 120 min, Example 1 - Residual double bond content: 0.6% — HNBR 4 at 120 min, Example 1 - Residual double bond content: 0.3% — HNBR 1 at 180 min, Example 1 - Residual double bond content: 0.3% — HNBR 2 at 180 min, Example 1 - Residual double bond content: 1.3% — HNBR 3 at 180 min, Example 1 - Residual double bond content: 0.3% — HNBR 4 at 180 min, Example 1 - Residual double bond content: 0.6% — HNBR 3 at 240 min, Example 1 - Mw: 191541 g/mol — HNBR 1, Example 1 - Mn: 68643 g/mol — HNBR 1, Example 1 - PDI: 2.79 — HNBR 1, Example 1 - Mooney viscosity: 38.2 — HNBR 1, Example 1 - Mw: 135772 g/mol — HNBR 2, Example 1 - Mn: 58259 g/mol — HNBR 2, Example 1 - PDI: 2.33 — HNBR 2, Example 1 - Mooney viscosity: 20.2 — HNBR 2, Example 1 - Mw: 195304 g/mol — HNBR 3, Example 1 - Mn: 64130 g/mol — HNBR 3, Example 1 - PDI: 3.05 — HNBR 3, Example 1 - Mooney viscosity: 69.3 — HNBR 3, Example 1 - Mw: 144212 g/mol — HNBR 4, Example 1 - Mn: 59851 g/mol — HNBR 4, Example 1 - PDI: 2.45 — HNBR 4, Example 1 - Mooney viscosity: 32.6 — HNBR 4, Example 1

Relevant Experimental Evidence - Example 1: Compared Grubbs III, Wilkinson's, and Grubbs II catalysts for simultaneous hydrogenation/metathesis and showed faster hydrogenation for Grubbs III at equal metal content. - Example 1: Demonstrated that the process can reduce residual double bonds to below 1% while also lowering Mw and Mooney viscosity. - Example 1: Showed that catalyst choice strongly affects final polymer properties, with HNBR 2 giving the lowest Mooney viscosity and HNBR 3 retaining the highest viscosity.

Technical Relevance This is one of the most directly relevant patents because it explicitly teaches simultaneous hydrogenation and metathesis of NBR to make HNBR in a single step. It provides concrete evidence for catalyst selection, residual unsaturation control, and molecular-weight reduction in the target material.

Patent 4

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a process for preparing low molecular weight HNBR by simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen and optionally a co-olefin, using ruthenium- or osmium-based

complexes. The invention emphasizes improved catalyst performance relative to prior art and aims to produce HNBR with lower Mw and narrower MWD in a single step. - Catalyst amount: 0.041 phr — HNBR 1, Example 1 - Catalyst amount: 0.045 phr — HNBR 2, Example 1 - Catalyst amount: 0.060 phr — HNBR 3, Example 1 - Metal content: 65.6 ppm — HNBR 1, HNBR 2, HNBR 3, Example 1 - Co-catalyst: NIL — HNBR 1, Example 1 - Co-catalyst: NIL — HNBR 2, Example 1 - Co-catalyst: triphenylphosphine — HNBR 3, Example 1 - Co-catalyst amount: 1.0 phr — HNBR 3, Example 1 - Reaction time: 0 min — Example 1 - Reaction time: 60 min — Example 1 - Reaction time: 120 min — Example 1 - Reaction time: 180 min — Example 1 - Reaction time: 240 min — Example 1 - Residual double bond content: 2.5% — HNBR 1 at 60 min, Example 1 - Residual double bond content: 7.0% — HNBR 2 at 60 min, Example 1 - Residual double bond content: 15.4% — HNBR 3 at 60 min, Example 1 - Residual double bond content: 0.6% — HNBR 1 at 120 min, Example 1 - Residual double bond content: 4.5% — HNBR 2 at 120 min, Example 1 - Residual double bond content: 3.1% — HNBR 3 at 120 min, Example 1 - Residual double bond content: 0.5% — HNBR 1 at 180 min, Example 1 - Residual double bond content: 3.9% — HNBR 2 at 180 min, Example 1 - Residual double bond content: 1.3% — HNBR 3 at 180 min, Example 1 - Residual double bond content: 0.4% — HNBR 1 at 240 min, Example 1 - Residual double bond content: 3.5% — HNBR 2 at 240 min, Example 1 - Residual double bond content: 0.7% — HNBR 3 at 240 min, Example 1 - Mw: 113371 g/mol — HNBR 1, Example 1 - Mn: 48352 g/mol — HNBR 1, Example 1 - PDI: 2.34 — HNBR 1, Example 1 - Mooney viscosity: 33.3 — HNBR 1, Example 1 - Mw: 133758 g/mol — HNBR 2, Example 1 - Mn: 54595 g/mol — HNBR 2, Example 1 - PDI: 2.45 — HNBR 2, Example 1 - Mooney viscosity: 41.3 — HNBR 2, Example 1 - Mw: 195304 g/mol — HNBR 3, Example 1 - Mn: 64130 g/mol — HNBR 3, Example 1 - PDI: 3.05 — HNBR 3, Example 1 - Mooney viscosity: 69.3 — HNBR 3, Example 1

Relevant Experimental Evidence - Example 1: Hoveyda and Arlt catalysts gave lower Mooney viscosity HNBR than Wilkinson's catalyst under the same metal content. - Example 1: The catalyst according to the invention achieved faster hydrogenation than the industrial benchmark catalyst. - Example 1: The process produced low-Mw HNBR with Mooney viscosity as low as 33.3 and residual double bond content below 1% after 240 min.

Technical Relevance This patent is highly relevant because it directly targets low molecular weight HNBR via simultaneous hydrogenation/metathesis, which is central to HNBR process innovation. It also provides comparative catalyst data showing how catalyst structure affects hydrogenation rate, residual unsaturation, and final viscosity.

Patent 5

- Patent Number: EP2292687B1
- Patent Title: ... containing ethylene acrylate (AEM), polyacrylate (ACM) and/or hydrogenated ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2012
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is primarily a vulcanizable-mixture patent, but it includes a detailed preparation example for HNBR by hydrogenating an NBR precursor in monochlorobenzene using Wilkinson's catalyst and triphenylphosphine at elevated temperature and hydrogen pressure. The HNBR is then used in rubber compounds with retarder systems and accelerators for improved scorch safety. - HNBR1 residual double bond content: 0.4% — Example 2 - HNBR1 maleic monoethylester content: 4.5% — Example 2 - HNBR1 Mooney viscosity (ML 1+4 @100°C): 101 — Example 2 - NBR1 butadiene content: 60.4% by weight — Table A, Example 2 - NBR1 maleic acid monoethyl ester content: 4.5% by weight — Table A, Example 2 - NBR1 acrylonitrile content: 35.1% by weight — Table A, Example 2 - NBR1 Mooney viscosity (ML 1+4 @100°C): 35 — Table A, Example 2 - NBR solution concentration: 12% total solids — Example 2 - Solvent: monochlorobenzene — Example 2 - Reaction temperature: 138 °C — Example 2 - Agitation speed: 600 rpm — Example 2 - Hydrogen pressure: 85 bar — Example 2 - Reaction time: 4 hrs — Example 2 - Co-catalyst: triphenylphosphine — Example 2 - MDR test temperature: 180 °C for 30' — Table 4, Example 2 - Stabaxol® I: 1.12 phr — Compound C, Example 1 - Benzoic acid: 0.70 phr — Compound D, Example 1 - Rhenogran® XLA-60: 2.00 phr — Compounds B, C, D, Example 1 - Rhenogran® DOTG-70: 2.85 phr — Compound A, Example 1 - Rhenogran® HMDC-70/AEMD: 0.85 phr — All compounds A-D, Example 1 - Press-curing temperature: 180 °C — Example 1 - Press-curing pressure: 100 bar — Example 1 - Post-curing temperature: 175 °C — Example 1 - Post-curing time: 4 hours — Example 1

Relevant Experimental Evidence - Example 1: Demonstrated that guanidine-free accelerator/retarder combinations can be formulated for HNBR vulcanization. - Example 2: Showed preparation of HNBR1 from NBR1 by hydrogenation at 138 °C and 85 bar H₂ for 4 hours. - Example 2: Showed that the resulting HNBR had very low residual double bond content (0.4%) and high Mooney viscosity (101), suitable for compounding.

Technical Relevance This patent is relevant because it contains an explicit HNBR hydrogenation preparation example and shows how the resulting HNBR is formulated into practical vulcanizable compounds. It is especially useful for understanding downstream processing requirements and the role of HNBR properties in cure behavior.

Patent 6

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenation of NBR to HNBR using ruthenium, palladium, or rhodium compounds, with a strong emphasis on catalyst residue removal and the resulting aging properties of the HNBR. The process is carried out in solution under elevated temperature and pressure, and the patent frames catalyst loading as a key variable for balancing rate, recovery, and product quality. - NBR repeat units: 40% to 90% by weight conjugated diene — Claim 1 - NBR repeat units: 10% to 60% by weight alpha,beta-unsaturated nitrile — Claim 1 - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200 °C — Claim 1 - Hydrogenation pressure: 700000 pascals to 15000000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - Hydrogenated HNBR aging criterion: gel content increase less than 10% after 4 days at 140 °C — Description - Hydrogenated HNBR aging criterion: Mooney viscosity increase less than 40% after 4 days at 140 °C — Description - EP-A-0482391 HNBR rhodium content: 58-60 ppm — Description - EP-A-0431370 purified HNBR rhodium content: 5.8 to 9.2 ppm — Description

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is directly relevant because it defines practical hydrogenation conditions and catalyst loadings for producing HNBR with improved aging properties. Its focus on catalyst residue control is important for industrial HNBR quality, color, and long-term stability.

Patent 7

- Patent Number: EP2519350B1
- Patent Title: Highly active metathesis catalysis selective for romp and rcm
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses novel ruthenium carbene complexes and their use in metathesis reactions, including depolymerization of NBR and subsequent hydrogenation to HNBR. The process descriptions include both depolymerization-first and hydrogenation-first variants, with hydrogenation performed under high pressure at elevated temperature. - Ligand synthesis scale: 5.0 mmol — Example 1 - Ligand synthesis solvent: DME 10

mL — Example 1 - Water: 3 mL — Example 1 - K₂CO₃: 1.5 eq — Example 1 - Reaction temperature: 85 °C — Example 1 - Pd(PPh₃)₄: 2% — Example 1 - Ligand 3a yield: 86% — Example 1 - Ru complex 4a yield: 4% — Example 1 - Ligand 3b synthesis scale: 5.0 mmol — Example 2 - Ligand 3b yield: 91% — Example 2 - Ru complex 4b yield: 29% — Example 2 - Ligand 3c synthesis scale: 5.0 mmol — Example 3 - Ligand 3c yield: 54% — Example 3 - Ru complex 4c yield: 5% — Example 3 - Ligand 3d synthesis scale: 5.0 mmol — Example 4 - Ligand 3d yield: 62% — Example 4 - Ru complex 4d yield: 32% — Example 4 - Ligand 3e yield: 35% — Example 5 - Ru complex 4e yield: 35% — Example 5 - Ligand 3f synthesis scale: 5.0 mmol — Example 6 - Ligand 3f yield: 63% — Example 6 - Ru complex 4f yield: 14% — Example 6 - Ligand 3g synthesis scale: 5.0 mmol — Example 7 - Ligand 3g yield: 56% — Example 7 - Ru complex 4g yield: 8% — Example 7 - Ligand 3h synthesis scale: 5.0 mmol — Example 8 - Ligand 3h yield: 32% — Example 8 - Ru complex 4h yield: 42% — Example 8 - Ligand 3j synthesis scale: 5.0 mmol — Example 9 - Ligand 3j yield: 93% — Example 9 - Ru complex 4j yield: 48% — Example 9 - Ligand 3k synthesis scale: 5.0 mmol — Example 10 - Ligand 3k yield: 83% — Example 10 - Ru complex 4k yield: 68% — Example 10 - Ligand 3m synthesis scale: 5.0 mmol — Example 11 - Ligand 3m yield: 49% — Example 11 - Ru complex 4m yield: 52% — Example 11 - Ligand 3n synthesis scale: 5.0 mmol — Example 12 - Ligand 3n yield: 45% — Example 12 - Ru complex 4n yield: 64% — Example 12 - Ligand 3p synthesis scale: 5.0 mmol — Example 13 - Ligand 3p yield: 67% — Example 13 - Ru complex 4p yield: 26% — Example 13 - Ligand 3q synthesis scale: 5.0 mmol — Example 14 - Ligand 3q yield: 46% — Example 14 - Ru complex 4q yield: 38% — Example 14 - Ligand 3r synthesis scale: 5.0 mmol — Example 15 - Ligand 3r yield: 44% — Example 15 - Ru complex 4r yield: 14% — Example 15 - Ligand 3s synthesis scale: 5.0 mmol — Example 16 - Ligand 3s yield: 51% — Example 16 - Ru complex 4s yield: 85% — Example 16 - Ligand 3t synthesis scale: 5.0 mmol — Example 17 - Ligand 3t yield: 38% — Example 17 - Ru complex 4t yield: 23% — Example 17 - Ligand 3u synthesis scale: 5.0 mmol — Example 18 - Ligand 3u yield: 28% — Example 18 - Ru complex 4u yield: 32% — Example 18 - Ligand 3v synthesis scale: 5.0 mmol — Example 19 - Ligand 3v yield: 81% — Example 19 - Ru complex 4v yield: 10% — Example 19 - Ligand 3w yield: 31% — Example 20 - Ru complex 4w yield: 31% — Example 20 - Depolymerization temperature: 30 to 100 °C — Description - Hydrogenation temperature: 60 to 150 °C — Description - Hydrogen pressure: high pressure — Description

Relevant Experimental Evidence - Example 1: Demonstrated synthesis of a Ru complex precursor and final Ru complex using DME/water, K₂CO₃, and Pd(PPh₃)₄ at 85 °C. - Examples 1-20: Showed a family of Ru complexes with varying ligand structures and isolated yields, supporting catalyst design for metathesis applications. - Description: Claimed use of the catalysts for NBR depolymerization and for HNBR formation via depolymerization followed by hydrogenation or hydrogenation-first sequences.

Technical Relevance This patent is relevant because it provides new Ru catalyst structures intended for metathesis and explicitly mentions depolymerization of NBR and hydrogenation to HNBR. It supports catalyst development for controlling molecular weight and enabling alternative HNBR synthesis routes.

Patent 8

- Patent Number: US10119045B2
- Patent Title: Electroconductive silver paste
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is primarily directed to electroconductive silver paste, but it includes a preparation example for NBR and a hydrogenated NBR (HNBR) used as the rubber binder. The HNBR is prepared by hydrogenation of NBR and then incorporated into conductive paste formulations to evaluate stretchability, resistance, and storage stability.

- Butadiene: 49 parts by weight — Preparation Example 1 - Acrylonitrile: 51 parts by weight — Preparation Example 1 - Deionized water: 270 parts by weight — Preparation Example 1 - Sodium dodecylbenzenesulfonate: 0.5 part by weight — Preparation Example 1 - Condensed product of sodium naphthalenesulfonate: 2.5 parts by weight — Preparation Example 1 - tert-Dodecyl mercaptan: 0.3 part by weight — Preparation Example 1 - Triethanolamine: 0.2 part by weight — Preparation Example 1 - Sodium carbonate: 0.1 part by weight — Preparation Example 1 - Bath temperature: 15 °C — Preparation Example 1 - Potassium persulfate: 0.3 part by weight — Preparation Example 1 - Potassium persulfate solution water: 19.7 parts by weight — Preparation Example 1 - Polymerization time: 20 hours — Preparation Example 1 - Hydroquinone: 0.5 part by weight — Preparation Example 1 - Hydroquinone solution water: 19.5 parts by weight — Preparation Example 1 - NBR content of nitrile group: 51% by weight — Example 3 - NBR alkali metal content: 50 ppm — Example 3 - NBR content of nitrile group: 51% by weight — Example 3, NBR (2) - NBR alkali metal content: 1400 ppm — Example 3, NBR (2) - NBR alkali metal content: 5500 ppm — Example 3, NBR (3) - NBR content of nitrile group: 36% by weight — Example 3, NBR (4) - NBR alkali metal content: 48 ppm — Example 3, NBR (4) - NBR content of nitrile group: 26% by weight — Example 3, NBR (5) - NBR alkali metal content: 52 ppm — Example 3, NBR (5) - HNBR content of nitrile group: 50% by weight — Example 3 - HNBR alkali metal content: 55 ppm — Example 3 - HNBR iodine value: 25 mg/100 mg — Example 3 - A/B ratio by weight: 81.5/18.5 — Example 3 - A/B ratio by weight: 77.4/22.6 — Example 3 - A/B ratio by weight: 89.4/10.6 — Example 3 - Specific resistance at 0% elongation: 1.5×10^{-4} ohm·cm — Example 3 - Specific resistance at 40% elongation: 3×10^{-4} ohm·cm — Example 3 - Specific resistance at 60% elongation: 6×10^{-4} ohm·cm — Example 3 - Specific resistance at 100% elongation: 15×10^{-4} ohm·cm — Example 3 - Specific resistance after 50 repetitive stretches: 2×10^{-4} ohm·cm — Example 3 - Specific resistance after 100 repetitive stretches: 4×10^{-4} ohm·cm — Example 3 - Specific resistance after 1000 repetitive stretches: 15×10^{-4} ohm·cm — Example 3 - Anti-migration property: 200 seconds — Example 3 - Storage stability: 25 °C for 3 months — Example 3 - Specific resistance at 0% elongation: 80×10^{-4} ohm·cm — Comparative Example 3 - Specific resistance at 40% elongation: 500×10^{-4} ohm·cm — Comparative Example 3 - Specific resistance at 60% elongation: 800×10^{-4} ohm·cm — Comparative Example 3 - Specific resistance at 100% elongation: 1500×10^{-4} ohm·cm — Comparative Example 3 - Anti-migration property:

Relevant Experimental Evidence - Preparation Example 1: Produced NBR by emulsion polymerization from butadiene and acrylonitrile using sodium dodecylbenzenesulfonate, sodium naphthalenesulfonate condensate, tert-dodecyl mercaptan, and potassium persulfate. - Example 3: Showed that HNBR with 50 wt% nitrile groups and 55 ppm alkali metal can be used in a conductive paste with good stretchability and low resistance. - Example 3: Demonstrated that lower alkali-metal content improves anti-migration and storage stability, while HNBR binder composition affects electrical performance.

Technical Relevance This patent is indirectly relevant because it uses HNBR as a functional binder and provides compositional evidence for hydrogenated nitrile rubber in a practical formulation. It is useful for understanding how HNBR quality parameters such as nitrile content, alkali-metal residue, and iodine value affect downstream performance.

Patent 9

- Patent Number: EP4215556A4
- Patent Title: LIQUID HYDROGENATED NITRILE BUTADIENE RUBBER, PROCESS THEREOF AND USE THEREOF
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent concerns liquid hydrogenated nitrile butadiene rubber and its process, but the extracted evidence does not include a detailed experimental synthesis example or explicit process conditions. The available text indicates that the invention is directed to hydrogenation of nitrile butadiene rubber and related catalyst systems, but the specific operational parameters are not disclosed in the provided evidence block.

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is relevant because it is explicitly directed to liquid HNBR, a specialized hydrogenated nitrile rubber form. However, the provided evidence does not expose enough process detail to extract concrete synthesis conditions or example outcomes.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US9079979B2 and EP2473281B1 both disclose simultaneous hydrogenation/metathesis routes for HNBR, but EP2473281B1 emphasizes Hoveyda/Arlt-type catalysts and lower final Mooney viscosity, while US9079979B2 compares Grubbs III, Grubbs II, and Wilkinson's catalyst performance.
- US9079979B2 and EP3827029A1 both use noble-metal hydrogenation chemistry, but EP3827029A1 focuses on catalyst residue control and aging stability with explicit Ru/Pd/Rh ppm windows, whereas US9079979B2 focuses on one-pot hydrogenation/metathesis and polymer-property outcomes.
- US8957161B2 and US7737233B2 are upstream metathesis-enabling patents rather than direct hydrogenation patents; both address catalyst activation for NBR degradation, with US8957161B2 using salt additives such as LiBr/CsBr and US7737233B2 using alkaline earth metal chlorides.
- EP2519350B1 extends the metathesis toolbox by introducing new Ru complexes and explicitly claims NBR depolymerization followed by hydrogenation to HNBR, bridging catalyst design and polymer processing.
- EP2292687B1 is formulation-oriented rather than synthesis-oriented, but it provides a concrete HNBR hydrogenation example at 138 °C and 85 bar H₂ and shows how HNBR properties translate into vulcanizable compounds.
- US10119045B2 is the least synthesis-focused of the set; it uses HNBR as a binder in conductive paste and highlights the importance of low alkali-metal content and controlled nitrile content for performance.
- Across the hydrogenation-focused patents, the recurring process window is elevated temperature, high hydrogen pressure, and ruthenium-based catalysts, with product quality strongly tied to residual double bond content, Mooney viscosity, and catalyst residue management.

4. CONCLUSION

The patent set shows two complementary routes to HNBR: direct hydrogenation of NBR and metathesis-assisted molecular-weight reduction followed by hydrogenation or simultaneous hydrogenation/metathesis. The strongest evidence supports ruthenium-based catalyst systems operated under elevated temperature and hydrogen pressure, with careful control of catalyst loading and residue removal to preserve aging performance and avoid discoloration or gel formation. For low-Mooney HNBR, the evidence favors preconditioning NBR to below about 30 Mooney units or using one-pot metathesis/hydrogenation systems that deliver Mw in the 20,000-250,000 g/mol range and residual double bonds below about 1%. Practical synthesis windows supported by the evidence include hydrogenation at roughly 60-200 °C and 700000-15000000 Pa, with catalyst loadings in the tens of ppm for noble metals. For process development, the most robust strategy is a ruthenium-catalyzed hydrogenation route with strong attention to catalyst recovery, or a metathesis-plus-hydrogenation route when lower molecular weight and narrower MWD are required.

5. REFERENCES

- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
- US7737233B2 | Catalyst systems and their use for metathesis reactions | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7737233B2/en>
- US9079979B2 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9079979B2/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- EP2292687B1 | ... containing ethylene acrylate (AEM), polyacrylate (ACM) and/or hydrogenated ... | Not disclosed | EP | 2012 | <https://patents.google.com/patent/EP2292687B1/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>
- EP2519350B1 | Highly active metathesis catalysis selective for romp and rcm | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2519350B1/en>
- US10119045B2 | Electroconductive silver paste | Not disclosed | US | 2018 | <https://patents.google.com/patent/US10119045B2/en>
- EP4215556A4 | LIQUID HYDROGENATED NITRILE BUTADIENE RUBBER, PROCESS THEREOF AND USE THEREOF | Zannan Scitech Co Ltd | EP | 2024 | <https://patents.google.com/patent/EP4215556A4/en>